

AI ETHICS LEARNING TOOLKIT

Is AI Sustainable?

Exploring Environmental Impacts

“Generative AI has a very disproportionate energy and carbon footprint with very little in terms of positive stuff for the environment.”

– Dr. Sasha Luccioni, Artificial Intelligence Researcher & Climate Lead, Hugging Face

The rapid growth of AI has environmental implications, raising concerns regarding its sustainability and carbon footprint. AI systems require vast amounts of computational power, which could lead to high energy consumption and greenhouse gas emissions. The data centers that support AI operations contribute to environmental degradation, water consumption, and e-waste. Research by leading AI sustainability expert Sasha Luccioni estimates that AI-generated text responses, such as those from Google AI Overview or ChatGPT, consume 30 times more energy to generate new text versus extracting text from a source. As AI continues to evolve, it is crucial to evaluate its environmental costs and explore ways to

mitigate its impact. Instructors may encourage students to be mindful of the environmental impact of AI as they explore its applications and reflect on the balance between convenience and sustainability.

Learning Activities



Conversation Starters A Few Questions to Get the Discussion Going

- In what ways do you think AI technologies impact the environment, both positively and negatively?
- Who should be responsible for making AI environmentally sustainable? Why?
- Can AI be made more eco-friendly? How?
- Have you seen/heard about examples of AI being used to help the environment?



Active Learning with AI Fun Ways to Explore AI's Strengths and Limitations

- In teams, students can use the [ChatUI Energy Bot](#) to investigate the energy consumption of everyday tasks when using AI. Students find one example of AI energy use and compare it to something familiar, like charging a phone.
- Students could track their carbon footprint in a log for a week

related to AI.

- **No AI Alternatives:**

- Students investigate how much water is used to cool AI data centers and compare it to daily household water use.
- Students research the difference in energy consumption per query between ChatGPT and Google and analyze the differences.



Disciplinary Extensions Ideas for Exploring AI's Impact in Specific Fields

- **Business/Markets & Management:** What is the responsibility of tech companies in reducing AI's environmental impact? Some companies have been pulling back on sustainability goals. Investigate what companies have done and how it compares to what they're currently doing.
- **Environmental Science:** AI's applications in climate modeling and conservation efforts versus its unintended consequences on natural resources.
- **Arts & Media Studies:** AI-generated images/media carry a particularly large carbon footprint. What are the ethics and implications of AI's enviro impact in visual/media arts?
- **Public Policy:** How awareness of AI's environmental impacts can inform regulation, energy standards, and national sustainability goals in the tech sector.

Resources

- Logan, C. [Against AI and Its Environmental Harms-keep | Riseup Pad](#). Excellent megathread/aggregator of

all kinds of AI and the enviro sources. Organized by source types. [Aggregated bibliography] 

- Luccioni, S., Trevelin, B. & M. Mitchell. (2024, September 24). [The Environmental Impacts of AI — Primer](#). [Article]. Excellent starting place. Luccioni and her team are leading scholars in AI and the environment. 
- [Cartography of generative AI](#). [Website] 

Scholarly

- Vincent, J. (2024, February 16). [How much electricity do AI generators consume?](#) *The Verge*. [Article] 
- Parshall, A. (2025, June 11). [What Do Google's AI Answers Cost the Environment?](#) *Scientific American*. [Magazine article] 
- Verma, P., & Tan, S. (2024, September 18). [A bottle of water per email: The hidden environmental costs of using AI chatbots](#). *Washington Post*. [News article] 
- Kwong, E. (2025, March 30). [AI and the Environment](#) NPR. Duration: 12 min. [Podcast]. 
- Dope Labs. (2025, May 11). [AI's environmental cost](#). Duration: 38 minutes. [Podcast]. 
- Roose, K., Newton, C., et al. (2025, January 17). [Goodbye TikTok, Ni Hao RedNote? + A.I.'s Environmental Impact + Meta's Masculine Energy](#). *The New York Times*. [Podcast] 

Recommendations

- Related topics → Do we need AI? Who builds our AI?
- [AI Pedagogy Project \(Harvard\) Assignments](#) → Filter by theme (e.g. misinformation) and/or subject (e.g. science & technology studies)
- [Sustainability-related Articles](#) from the AI Ethics & Policy News Aggregator sourced by [Casey Fiesler](#).
Note: This would be an excellent place to identify recent news stories you could share with students, or incorporate into a case study.

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1. Hao, K. (2025). *Empire of AI: Dreams and Nightmares in Sam Altman's OpenAI*. Penguin Press. Pg. 276
 2. Luccioni, S., Jernite, Y., & Strubell, E. (2024). Power Hungry Processing: Watts Driving the Cost of AI Deployment? *The 2024 ACM Conference on Fairness, Accountability, and Transparency*, 85–99.
<https://doi.org/10.1145/3630106.3658542>

PHYSICAL LOCATION

The Erwin Square Mill Building
2024 West Main Street
Durham, NC 27705

MAILING ADDRESS

Box 90700
Durham, NC 27708

CONTACT US

Phone: (919) 660-5806
Email:
learninginnovation@duke.edu

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